

- Q.26 Prepare a simple part program for turning.
 Q.27 Explain part program structure with example.
 Q.28 Give the comparison between Ne and CNC machines.
 Q.29 Enlist the five applications of Ultrasonic machining.
 Q.30 Enlist the function of Dielectric fluid used in electric discharge machining.
 Q.31 Write different types of milling operation in CNC milling.
 Q.32 Explain the working principles of Electron Beam Machining.
 Q.33 Describe the primary motions of robots.
 Q.34 Name any five industrial applications of Robots.
 Q.35 Describe the robot configurations.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
 Q.36 Explain the working principle of Ultrasonic machining process. Give its advantages and limitations also.
 Q.37 Enumerate ten applications of CNC machines.
 Q.38 Explain any five robotic sensors.

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5th Sem / Auto Sub : Advanced Manufacturing Processes

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Frequency of tool's oscillation in USM ranges between
 a) 5-10 kHz b) 10-15 kHz
 c) 18-20 kHz d) 25-50 kHz
- Q.2 NC contouring is an example of
 a) Continuous path positioning
 b) Point-Point positioning
 c) Absolute positioning
 d) Incremental positioning
- Q.3 What does CNC stands for
 a) Computer Network Control
 b) Computer Numerical Control
 c) Computer Nested Control
 d) Computer Numerical Controller
- Q.4 Command G00 means
 a) Feed motion in a circular path
 b) Rapid Motion
 c) Feed motion in straight line
 d) Slow Motion

- Q.5 A CNC machines controls the movement of the machine to
- Automate
 - Monitor
 - Both A & B
 - None
- Q.6 In a CNC Machine a command
- Start with a numerical value and ends with a alphabet
 - Start with a numerical value and ends with two alphabets
 - Start with a alphabet and ends with a numerical value
 - None
- Q.7 In USM, material removal occurs due to
- Abrasion action
 - Corroding action
 - Chemical action
 - Chip formation
- Q.8 What is the full form of LBM in advanced machining processes?
- Laser Beam Manufacturing
 - Laser Beam Machining
 - Light Blast Manufacturing
 - Light Beam Machining
- Q.9 Ultrasonic machining is based upon
- Vibrational waves of zero frequency
 - Vibrational waves of low frequency
 - Vibrational waves of high & low frequency
 - Vibrational waves of high frequency
- Q.10 Which is not the type of part programming format?
- Fixed block format
 - Variable block format
 - Tab sequential format
 - Word address format

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 _____ gas is filled in flash lamp is laser beam machining.
- Q.12 _____ code is used for cutter radius compensation right.
- Q.13 Name the M code for "Spindle Stop" and "Tool Change".
- Q.14 Give the use of collet in milling machine.
- Q.15 Name any one type of tape reader.
- Q.16 Write the function of Machine control Unit.
- Q.17 G04 code is used to specify_____.
- Q.18 Down milling is also called_____.
- Q.19 What is the other name of electric discharge machining?
- Q.20 Name a work holding device in milling.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write any five advantages and disadvantages of DNC system.
- Q.22 Differentiate between absolute coordinate system and incremental coordinate system
- Q.23 Prepare a simple part program for drilling.
- Q.24 Describe circular interpolation with example.
- Q.25 Describe any five M code in part programming.